Saakash Sathyan

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Education

University of California, Irvine Expected June 2027

B.S. Aerospace Engineering

- Relevant Coursework: Dynamics, Fluid Dynamics I-III, Thermodynamics, Astronautics, Circuits, Materials Science, Aerodynamics, Mechanics of Structures, Mechanical Systems Lab
- Relevant Organizations: American Institute of Aeronautics and Astronautics (AIAA), Students for the Exploration and Development of Space (SEDS)

Projects

UCI MAE 106 Autonomous Pneumatic Robot | *Mechanical Coordinator* Mar 2026 – Present

- Designed full chassis assembly in SolidWorks including frame, wheel mounts, and steering linkage
- Engineered pneumatic propulsion system harnessing a single cylinder powered by a compressed air tire
- Designed steering mechanism actuated by servo motor to enable autonomous directional control
- Ensured compliance with competition constraints: 12" diameter envelope and 16" air tank tire height
- Fabricated components using band saw, drill press, laser cutter, and 3D printer with PLA filament
- Collaborated with Electrical and Controls coordinators to integrate pneumatic, servo, and Arduino systems
- Robot to compete autonomously using bearing-based orientation and wheel-rotation odometry for navigation

SEDS @ UCI Autonomous Mars Rover | *Fabrication Subteam* Nov 2025 – Present

- Designed autonomous rover to traverse uneven terrain and retrieve a rock sample without human input
- Modeled full chassis assembly in SolidWorks including dual-deck frame, 4 motor mounts, and wheel hubs
- Developed a 1:1 scaled electronics bay layout (30 × 20 cm) for electrical subteam coordination
- Integrated 3 subassemblies (turning mechanism, 2-DOF arm, claw) into the full rover assembly
- Adapted drivetrain from 4-wheel to rear-wheel drive following parts confirmation
- Selected plywood chassis decks and FDM 3D printed components for arm, claw, and turning mechanism
- Created system-level concept sketches defining rover architecture and component relationships across 3 subteams
- Utilized Orca Slicer and Bambu X1C 3D printers to fabricate arm, claw, and turning components

Leadership & Activities

Financial Director Nov 2025 – Present

AIAA @ UCI *Irvine, CA*

- Oversee budget planning and financial reporting for a ~$10,000 annual budget
- Manage financial operations supporting a 100+ member student engineering organization
- Personally managed 2 fundraising events from planning through execution and financial reconciliation
- Coordinate sponsorship acquisition and develop proposals for aerospace industry partners
- Support organizational operations and strategic planning across multiple concurrent events
- Track and reconcile spending across all chapter activities to ensure responsible use of funds

Certifications & Training

UAV Design and Analysis Virtual Internship April 2026 - Present

Skyline Space *India - Remote*

- Completing coursework in UAV aerodynamics, flight dynamics, stability analysis, and autonomous systems
- Utilizing SolidWorks and ANSYS software to model drone components and perform CFD simulations

Technical Skills

Programming: MATLAB, Python, C++, RobotC
CAD & Design: SolidWorks, AutoCAD, Fusion 360, OnShape, Inventor, Parametric Modeling, Orca Slicer, Bambu Studio
Engineering: Mechanical Design, CAD Modeling, System Integration, Component Assembly, Testing, Aerodynamic Analysis, Technical Documentation, Design Review
Fabrication & Hardware: Hand & Power Tools, Band Saw, Table Saw, Belt Sander, FDM 3D Printing, Soldering, Oscilloscope, Function Generator, Digital Calipers
Productivity: Microsoft Office, Google Suite, Adobe Photoshop